American Computer Science League

2022-2023 ● Contest 1: Short Problems ● Classroom Division

1. Computer Number Systems

Convert 2023_{10} to octal.

A. 2023
B. 3747
C. 4040
D. 4511
E. 4737

2. Computer Number Systems

What is the value of $657383_{16} - 654321_{16}$ in base 8?

A. 30242
B. 31142
C. 30152
D. 30142
E. 30132

3. Recursive Functions

Find $f(18)$ given:

$$f(x) = \begin{cases} f(x-3) + 1 & \text{if } x \geq 10 \\ f(x-2) - 3 & \text{if } 5 < x < 10 \\ x^2 & \text{if } x \leq 5 \end{cases}$$

A. 18
B. 19
C. 20
D. 21
E. 22

4. Recursive Functions

Find $f(12, 21)$ given:

$$f(x,y) = \begin{cases} f(x-1, y+2) - 3 & \text{if } x > y \\ f(y-1, x+2) + 1 & \text{if } x < y \\ x & \text{if } x = y \end{cases}$$

A. 13
B. 18
C. 20
D. 15
E. 12

5. What Does This Program Do? (Branching)

What is the output for this program given the following inputs:
a = 4, b = 5, c = 7

```
input a, b, c
if a^2 + b^2 == c^2 then
     ans = a * b / 2
else
     s = (a + b + c) / 2
     ans = sqrt(s * (s-a) * (s-b) * (s-c))
end if
output int(ans)
```

A. 96
B. 9
C. 10
D. 9.8
E. 14

6. Computer Number Systems

How many 1s are there in the binary representation of the decimal numbers from 8 to 16, inclusive?

A. 23
B. 21
C. 19
D. 17
E. 15

7. Computer Number Systems

Given the following 5 numbers:

$78B_{16}$ ACE_{16} $F4A_{16}$ $2EF_{16}$ BAD_{16}

Which one has the most odd numbers in its octal equivalent?

A. 78B
B. ACE
C. F4A
D. 2EF
E. BAD

8. Recursive Functions

Find $f(25)$ given:

$$f(x) = \begin{cases} 2 & \text{if } x = 1 \\ 4 & \text{if } x = 2 \\ 2 \cdot f(x-1) / f(x-2) & \text{if } x \geq 3 \end{cases}$$

A. 2
B. 4
C. 1
D. 6
E. 3

9. Recursive Functions

Find $f(f(f(9)))$ given:

$$f(x) = \begin{cases} f(x-2) + 2 & \text{if } x \text{ is a multiple of } 4 \\ & \quad \text{and not a multiple of } 3 \\ f(x+1) - 1 & \text{if } x \text{ is a multiple of } 3 \\ & \quad \text{and not a multiple of } 4 \\ x - 3 & \text{otherwise} \end{cases}$$

A. 7
B. 6
C. 4
D. 3
E. 0

10. What Does This Program Do? (Branching)

Which of the following values for a, b, and c will output OBTUSE and SCALENE?

```
input a, b, c
if c >= a + b then
    output "NOT A TRIANGLE"
else
    if a ^ 2 + b ^ 2 == c ^ 2 then
        output "RIGHT"
    else if a ^ 2 + b ^ 2 > c ^ 2 then
        output "ACUTE"
    else
        output "OBTUSE"
    end if
    if a == b and a == c then
        output "EQUILATERAL"
    end if
    if a == b or a == c or b == c then
        output "ISOSCELES"
    else
        output "SCALENE"
    end if
end if
```

A. a=3, b=6, c=9
B. a=4, b=5, c=6
C. a=5, b=6, c=10
D. a=6, b=6, c=6
E. a=6, b=6, c=10